



Operating System Lab Set-Up

ENCS303

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Environment Setup Verification Report

Windows 11 → WSL2 → Ubuntu 24.04 LTS → VS Code → Python Virtual Environment → Lab Structure

Student system: Windows 11, WSL2 | **Linux user:** titan | **Lab folder:** D:\Operating-Systems-Lab (/mnt/d/Operating-Systems-Lab)

Reference guide: Operating Systems Laboratory - Complete Environment Setup Guide (os_lab_tutorial.pdf)

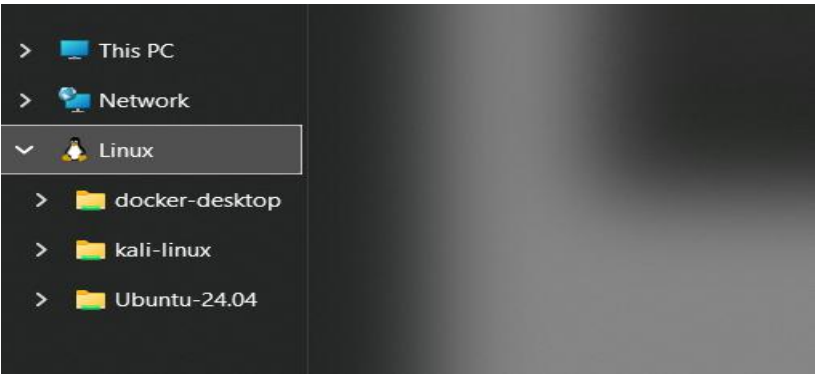
1. Platform & Distribution

```
akshay_yadav_ubutu@Akshayyadavpc:~$ cd Operating-Systems-Lab
akshay_yadav_ubutu@Akshayyadavpc:~/Operating-Systems-Lab$ ls
Capstone Lab1 Lab2 Lab3 Lab4 README.md common mkdir requirements.txt
akshay_yadav_ubutu@Akshayyadavpc:~/Operating-Systems-Lab$
```

Ubuntu (WSL2) terminal running on Windows 11

```
PS C:\Users\aksha> wsl -l -v
NAME                STATE              VERSION
* docker-desktop    Stopped            2
kali-linux           Stopped            2
Ubuntu-24.04        Running             2
PS C:\Users\aksha>
```

```
PRETTY_NAME="Ubuntu 24.04.4 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
VERSION="24.04.4 LTS (Noble Numbat)"
VERSION_CODENAME=noble
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo
```



Windows Explorer showing installed WSL Linux distributions

2. Installed Command – Line Tools

```
akshay_yadav_ubutu@Akshayyadavpc:~$ python3 --version
Python 3.12.3
akshay_yadav_ubutu@Akshayyadavpc:~$ pip --version
pip 24.0 from /usr/lib/python3/dist-packages/pip (python 3.12)
akshay_yadav_ubutu@Akshayyadavpc:~$ git --version
git version 2.43.0
akshay_yadav_ubutu@Akshayyadavpc:~$ bash --version
GNU bash, version 5.2.21(1)-release (x86_64-pc-linux-gnu)
Copyright (C) 2022 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>

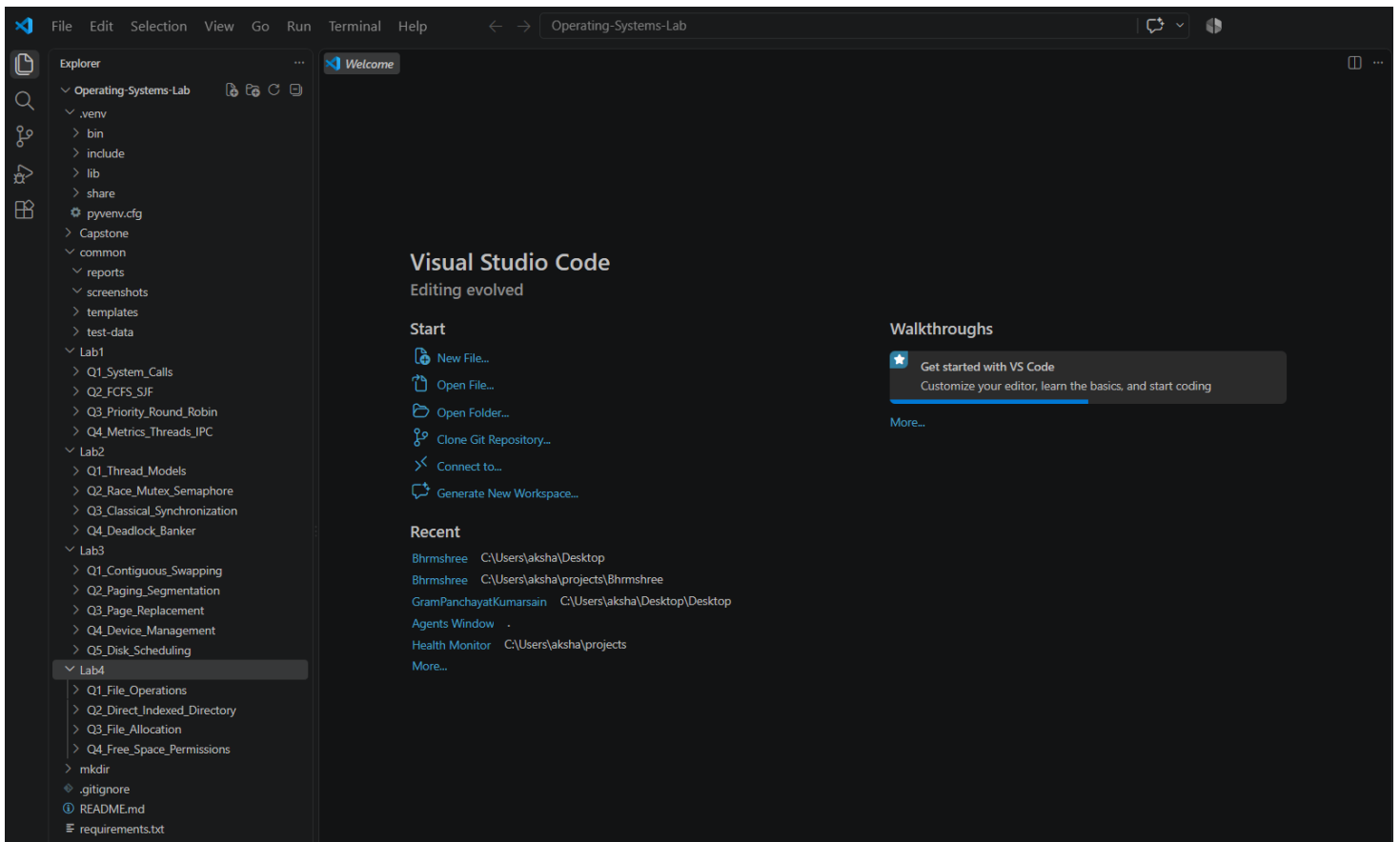
This is free software; you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
akshay_yadav_ubutu@Akshayyadavpc:~$ htop --version
htop 3.3.0
akshay_yadav_ubutu@Akshayyadavpc:~$ tree --version
tree v2.1.1 © 1996 - 2023 by Steve Baker, Thomas Moore, Francesc Rocher, Florian Sesser, Kyosuke Tokoro
akshay_yadav_ubutu@Akshayyadavpc:~$ iostat -V
sysstat version 12.6.1
(C) Sebastien Godard (sysstat <at> orange.fr)
akshay_yadav_ubutu@Akshayyadavpc:~$ curl --version
curl 8.5.0 (x86_64-pc-linux-gnu) libcurl/8.5.0 OpenSSL/3.0.13 zlib/1.3 brotli/1.1.0 zstd/1.5.5 libidn2/2.3.7 libpsl/0.21.2 (+libidn2/2.3.7) libssh/0.10.6/openssl/zlib nghttp2/1.59.0 librtmp/2.3 OpenLDAP/2.6.10
Release-Date: 2023-12-06, security patched: 8.5.0-2ubuntu10.11
Protocols: dict file ftp ftps gopher gophers http https imap imaps ldap ldaps mqtt pop3 pop3s rtmp rtsp scp sftp smb smbs smtp smtps t
elnet tftp
Features: alt-svc AsynchDNS brotli GSS-API HSTS HTTP2 HTTPS-proxy IDN IPv6 Kerberos Largefile libz NTLM PSL SPNEGO SSL threadsafe TLS-
SRP UnixSockets zstd
akshay_yadav_ubutu@Akshayyadavpc:~$ █
```

Tools	Required for	Status
Python3	Runs all Python lab programs	INSTALLED
Pip3	Package install / virtual env	INSTALLED
Git	Version control	INSTALLED
htop	Process & memory monitoring	INSTALLED
tree	Folder-tree display	INSTALLED
sysstat	I/O performance stats	INSTALLED
curl	Network utility	INSTALLED

3. Linux OS Command Tests

```
akshay_yadav_ubutu@Akshayyadavpc: $ ps -ef | head -6
UID          PID    PPID  C  STIME TTY          TIME CMD
root           1        0  0   14:31 ?        00:00:00 /sbin/init
root           2        1  0   14:31 ?        00:00:00 /init
root           6        2  0   14:31 ?        00:00:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root          42        1  0   14:31 ?        00:00:00 /usr/lib/systemd/systemd-journald
root          89        1  0   14:31 ?        00:00:01 /usr/lib/systemd/systemd-udevd
akshay_yadav_ubutu@Akshayyadavpc: $ free -h
               total        used        free      shared  buff/cache   available
Mem:           7.4Gi        554Mi        6.8Gi        3.5Mi        184Mi        6.9Gi
Swap:          2.0Gi          0B          2.0Gi
akshay_yadav_ubutu@Akshayyadavpc: $ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            3.7G  0  3.7G  0% /usr/lib/modules/6.6.87.2-microsoft-standard-WSL2
none            3.7G  4.0K  3.7G  1% /mnt/wsl
drivers         925G  266G  660G  29% /usr/lib/wsl/drivers
/dev/sdd        1007G  2.2G  954G  1% /
none            3.7G   80K  3.7G  1% /mnt/wslg
none            3.7G   0  3.7G  0% /usr/lib/wsl/lib
rootfs          3.7G  2.7M  3.7G  1% /init
none            3.7G  528K  3.7G  1% /run
none            3.7G   0  3.7G  0% /run/lock
none            3.7G   0  3.7G  0% /run/shm
none            3.7G  100K  3.7G  1% /mnt/wslg/versions.txt
none            3.7G  100K  3.7G  1% /mnt/wslg/doc
C:\             925G  266G  660G  29% /mnt/c
tmpfs           758M   20K  758M  1% /run/user/1000
```

4. Visual Studio Code



5. Python Virtual Environment & Libraries

```
akshay_yadav_ubutu@Akshayyadavpc:~/Operating-Systems-Lab$ code .
akshay_yadav_ubutu@Akshayyadavpc:~/Operating-Systems-Lab$ source .venv/bin/activate
(.venv) akshay_yadav_ubutu@Akshayyadavpc:~/Operating-Systems-Lab$ which python
/home/akshay_yadav_ubutu/Operating-Systems-Lab/.venv/bin/python
(.venv) akshay_yadav_ubutu@Akshayyadavpc:~/Operating-Systems-Lab$ python --version
Python 3.12.3
(.venv) akshay_yadav_ubutu@Akshayyadavpc:~/Operating-Systems-Lab$ python -c "import psutil, pandas, matplotlib; print('PYTHON LAB ENVIRONMENT READY!')"
PYTHON LAB ENVIRONMENT READY
```

6. Final Combined Readiness Test

```
(.venv) akshay_yadav_ubutu@Akshayyadavpc:~/Operating-Systems-Lab$ python -c "import os, sys, psutil, pandas, matplotlib;
(.venv) akshay_yadav_ubutu@Akshayyadavpc:~/Operating-Systems-Lab$ python -c "import os, sys, psutil, pandas, matplotlib;
(.venv) akshay_yadav_ubutu@Akshayyadavpc:~/Operating-Systems-Lab$ "
```

```
.
├── .gitignore
├── Capstone
│   ├── README.md
│   ├── data
│   ├── logs
│   ├── outputs
│   ├── screenshots
│   ├── src
│   └── tests
├── Lab1
│   ├── Q1_System_Calls
│   │   ├── README.md
│   │   ├── input
│   │   ├── output
│   │   ├── screenshots
│   │   ├── src
│   │   └── tests
│   ├── Q2_FCFS_SJF
│   │   ├── README.md
│   │   ├── input
│   │   ├── output
│   │   ├── screenshots
│   │   ├── src
│   │   └── tests
│   ├── Q3_Priority_Round_Robin
│   │   ├── README.md
│   │   ├── input
│   │   ├── output
│   │   ├── screenshots
│   │   ├── src
│   │   └── tests
│   └── Q4_Metrics_Threads_IPC
│       ├── README.md
│       ├── input
│       ├── output
│       ├── screenshots
│       ├── src
│       └── tests
├── Lab2
│   ├── Q1_Thread_Models
│   │   ├── README.md
│   │   ├── input
│   │   ├── output
│   │   ├── screenshots
│   │   ├── src
│   │   └── tests
│   └── Q2_Race_Mutex_Semaphore
│       ├── README.md
```

Environment setup and verification session log

7. Lab Folder Structure & Checklist

```
(.venv) aksnay_yadav_ubuntu@aksnayyadavpc: /operating-systems-lab$ tree -a -L 2 -I '.venv|__pycache__'
.
├── .gitignore
├── Capstone
│   ├── README.md
│   ├── data
│   ├── logs
│   ├── outputs
│   ├── screenshots
│   ├── src
│   └── tests
├── Lab1
│   ├── Q1_System_Calls
│   ├── Q2_FCFS_SJF
│   ├── Q3_Priority_Round_Robin
│   └── Q4_Metrics_Threads_IPC
├── Lab2
│   ├── Q1_Thread_Models
│   ├── Q2_Race_Mutex_Semaphore
│   ├── Q3_Classical_Synchronization
│   └── Q4_Deadlock_Banker
├── Lab3
│   ├── Q1_Contiguous_Swapping
│   ├── Q2_Paging_Segmentation
│   ├── Q3_Page_Replacement
│   ├── Q4_Device_Management
│   └── Q5_Disk_Scheduling
├── Lab4
│   ├── Q1_File_Operations
│   ├── Q2_Direct_Indexed_Directory
│   ├── Q3_File_Allocation
│   └── Q4_Free_Space_Permissions
├── README.md
├── common
│   ├── reports
│   ├── screenshots
│   ├── templates
│   └── test-data
├── mkdir
└── requirements.txt

35 directories, 4 files
```

Requirement	Verification	Status
WSL2 available	wsl -l -v	DONE
Ubuntu 24.04 LTS installed	cat /etc/os-release	DONE
Linux username created	whoami -> titan	DONE
Ubuntu packages updated & tools installed	apt install output	DONE
Python, pip, venv available	version checks	DONE
Git installed	git --version	DONE
Process / memory / disk tools available	ps, free, df, iostat	DONE
VS Code installed + WSL & Python extensions	code --list-extensions	DONE
Main laboratory folder created	D:\Operating-Systems-Lab	DONE
Virtual environment created & activated	.venv, (.venv) prompt	DONE
psutil, pandas, matplotlib installed	import test	DONE
Lab1-4 + Capstone structure created	tree output	DONE
Final Python readiness test passed	OPERATING SYSTEMS LAB SETUP READY	DONE